

Use-Case Flow Specification

Faculty Research Grant & Publication Tracker — Lab 1(PESIUG24CS497)

Use Case: UC-01 → Log Grant Expense

Primary Actor: Faculty Researcher

Supporting Actors: Research Dean (approves overdrafts)

Preconditions: Faculty Researcher is authenticated in the portal and is listed as an investigator on an active, funded grant.

Postconditions: The expense is recorded in the audit ledger, the grant's remaining budget balance is updated, and a confirmation is shown to the Faculty Researcher.

Main Success Scenario

1. Faculty Researcher selects "Log Expense" for a specific grant from the portal dashboard.
2. System displays a form requesting expense category (e.g., equipment, travel, supplies), amount, and a short description.
3. Faculty Researcher enters the expense details and submits the form.
4. System validates the inputs (amount is numeric and greater than zero, category is selected, grant is active).
5. System checks the current budget balance for the grant (includes UC-02: Check Budget Balance).
6. System confirms the expense amount does not exceed the remaining budget balance.
7. System records the expense entry in the immutable audit ledger and deducts the amount from the remaining balance.
8. System displays a confirmation message showing the new remaining balance.
9. Use case ends successfully.

Alternate Flow

Expense Exceeds Remaining Budget (extends UC-03: Approve Budget Overdraft)

- 6a1. System flags the expense entry as "Pending Dean Approval" and notifies the Research Dean.
- 6a2. Research Dean reviews the requested expense against the grant's remaining balance and either approves or rejects the overdraft.
- 6a3. If approved, the system posts the expense to the ledger, updates the balance (which may go negative), and notifies the Faculty Researcher. Use case resumes at step 7.
- 6a4. If rejected, the system discards the expense entry, notifies the Faculty Researcher with the reason, and the use case ends without posting the expense.